Table 6: Optimal learning rates for different models, task types, and localizations.

Model	64 athletes to random sport	Basketball Athletes to Golf	16 CounterFact facts	64 CounterFact facts
Gemma				
FLU	1×10^{-5}	2×10^{-5}	5×10^{-6}	1×10^{-5}
Localized CT	1×10^{-5}	5×10^{-6}	5×10^{-6}	2×10^{-5}
Nonlocalized	2×10^{-6}	5×10^{-6}	5×10^{-6}	2×10^{-6}
Gemma 2				
FLU	1×10^{-5}	5×10^{-5}	5×10^{-5}	2×10^{-5}
Localized CT	5×10^{-5}	5×10^{-5}	5×10^{-5}	2×10^{-5}
Nonlocalized	2×10^{-5}	1×10^{-5}	5×10^{-6}	5×10^{-6}
Llama 3				
FLU	5×10^{-5}	5×10^{-5}	1×10^{-4}	5×10^{-5}
Localized CT	1×10^{-4}	5×10^{-5}	2×10^{-5}	5×10^{-5}
Nonlocalized	2×10^{-5}	1×10^{-5}	2×10^{-5}	1×10^{-5}

Table 7: Optimal inject loss coefficients for different models, task types, and localizations.

Model	64 athletes to random sport	Basketball Athletes to Golf	16 CounterFact facts	64 CounterFact facts
Gemma				
FLU	5.0	5.0	0.1	2.0
Localized CT	0.1	1.0	0.2	1.0
Nonlocalized	0.2	1.0	0.2	1.0
Gemma 2				
FLU	1.0	5.0	2.0	0.5
Localized CT	5.0	5.0	2.0	2.0
Nonlocalized	5.0	5.0	1.0	1.0
Llama 3				
FLU	5.0	0.2	1.0	0.2
Localized CT	2.0	1.0	0.1	2.0
Nonlocalized	2.0	0.1	2.0	0.2

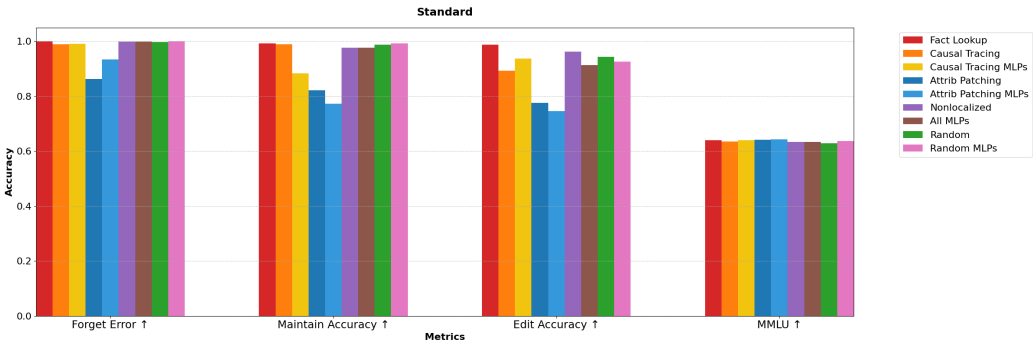

Figure 26: Standard Prompting results for Sports-Athlete-Editing, across all localizations.

A.7.3 Details on Adversarial Relearning

We retrain the model for 20 iterations with cross-entropy on half of the forget set (along with a standard retain and SFT loss), adding up all losses with loss coefficient 1.

We present relearning results for all localizations averaged over models. As shown in Figure 34, the FLU localization remains optimal, although the baselines of Nonlocalized, All MLPs, Random, and Random MLPs are competitive.

We also present relearning results on the other tasks. As mentioned in Section 3.2, since Full-Sports-Editing forget facts are not independent, we don't expect valid results from relearning. Thus, in Figure 35, we see that every localization regains 100% editing accuracy.

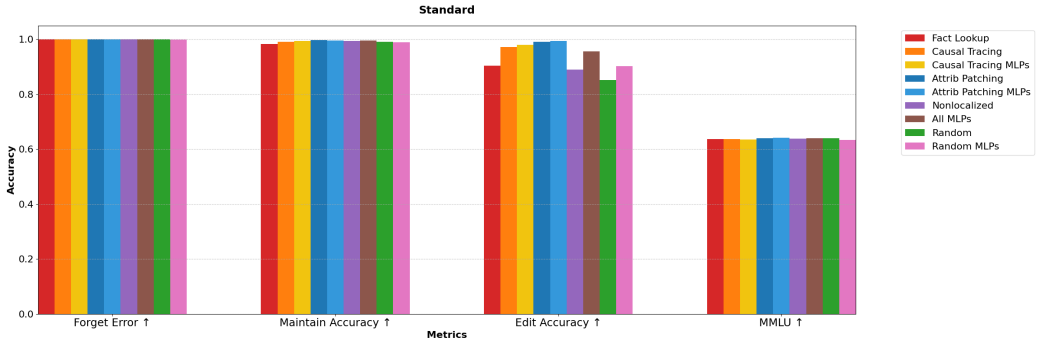

Figure 27: Standard Prompting results for Full-Sports-Editing, across all localizations.

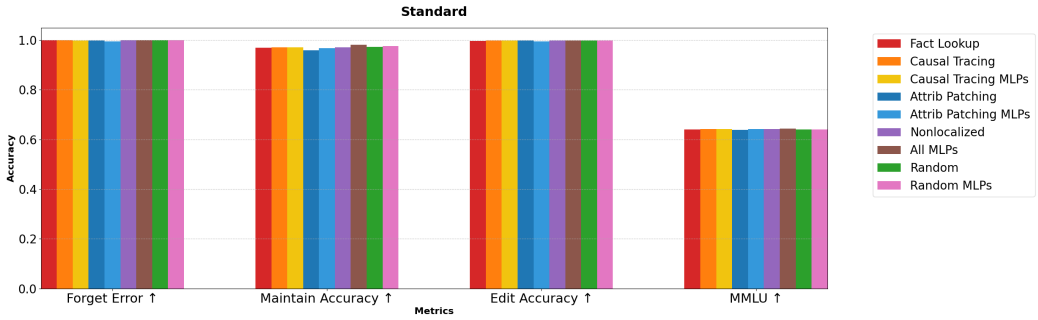

Figure 28: Standard Prompting results for CounterFact-Editing, across all localizations.

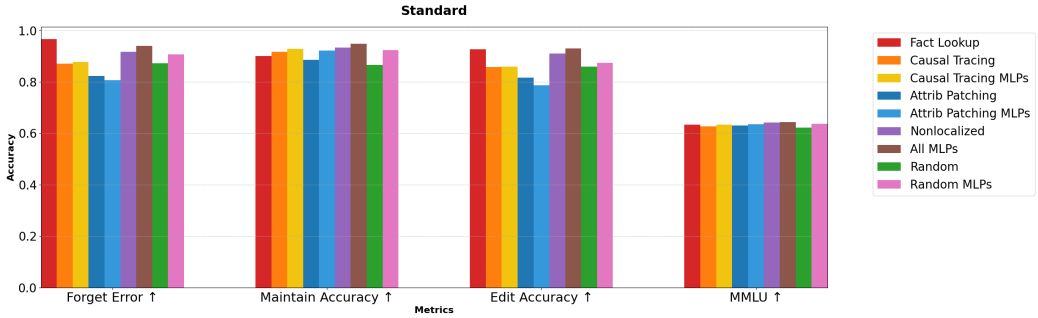

Figure 29: Standard Prompting results for Sequential-CounterFact-Editing, across all localizations.

On CounterFact-Editing and Sequential-CounterFact-Editing, as shown in Figures 36 and 37, none of the localizations relearn more than 7% accuracy, suggesting adversarial relearning was not a sufficiently strong enough evaluation for these tasks. Regardless, FLU localization is either the most or second-most robust localization to relearning, although localizations don't differ by much.

A.7.4 Details on Latent Knowledge

We present the probing classification accuracies for the three models separately here, as well as for all localizations we previously left out.

In Gemma-7b and Llama-3-8b, FLU probing is the most monotonic and the best in the early layers, either steadily decreasing to 0 for forget accuracy or increasing to 1 for edit accuracy, with the least extreme peaks. In Gemma-2-9b, the Nonlocalized, All MLPs, and Random MLPs baselines are competitive with FLU. The other OT localization, Attribution Patching, has 100% probing forget accuracy across many layers, suggesting it represents the ground truth answer very clearly.

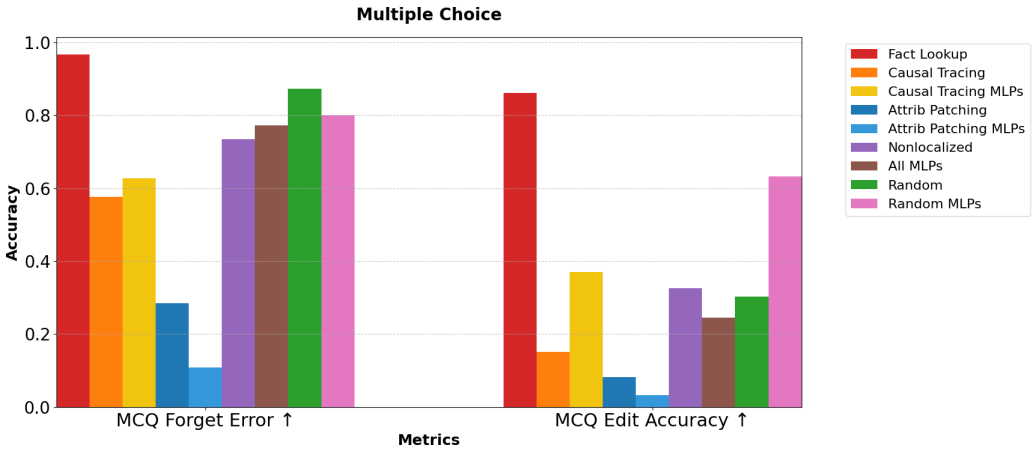

Figure 30: Adversarial Prompting results for Sports-Athlete-Editing, across all localizations.

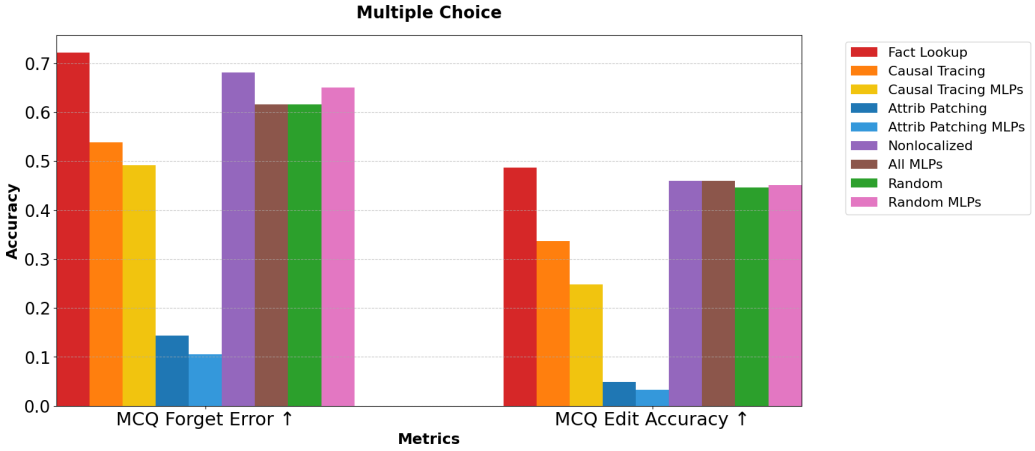

Figure 31: Adversarial Prompting results for Full-Sports-Editing, across all localizations.

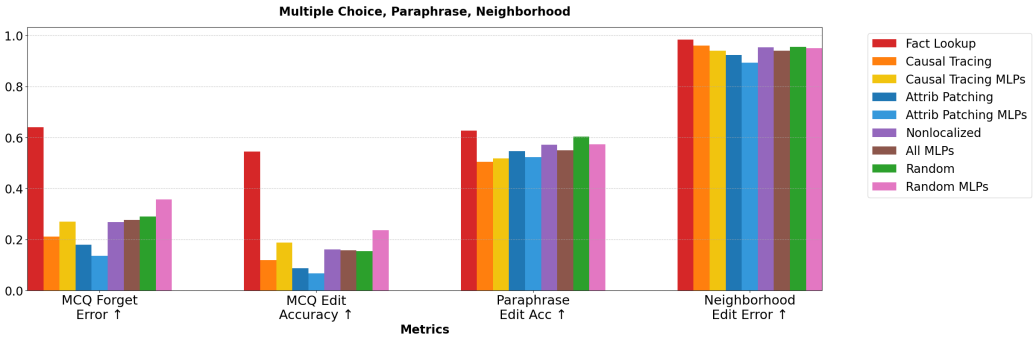

Figure 32: Adversarial Prompting results for CounterFact-Editing, across all localizations.

A.8 Soft Prompt Evaluations

Because many localizations seem to be weak to prompting schemes, we attempt a simple adaptive attack of soft prompts, where we optimize the continuous embeddings at the end of the prompt to recover the correct answer on half of our forget set. We then evaluate the model's performance on the other half, with this soft prompt in place (Lester et al., 2021). We average over evaluations from